

Qatar-1b

R-2747

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160712 · created 2026-09-02T22:56:02+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457581.70778 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1706 +/- 0.0097
Transit depth (Rp/Rs)^2	2.91 +/- 0.33 [%]
Orbital Inclination (inc)	85.95 +/- 1.19
Airmass coefficient 1 (a1)	1.3 +/- 0.016
Airmass coefficient 2 (a2)	-0.206 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	None
Optimal Aperture	3.69
Optimal Annulus	17.71
Transit Duration (day)	0.0783 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1706 $\pm$ 0.0097 vs 0.1463 (deviation 2.48 σ)
Duration fit vs published	0.0783 d vs 0.0692 d (deviation 2.26 σ)
Mid-time O-C	2.1 min
Depth SNR	17.4
Residual scatter	1.26 %

Light curve

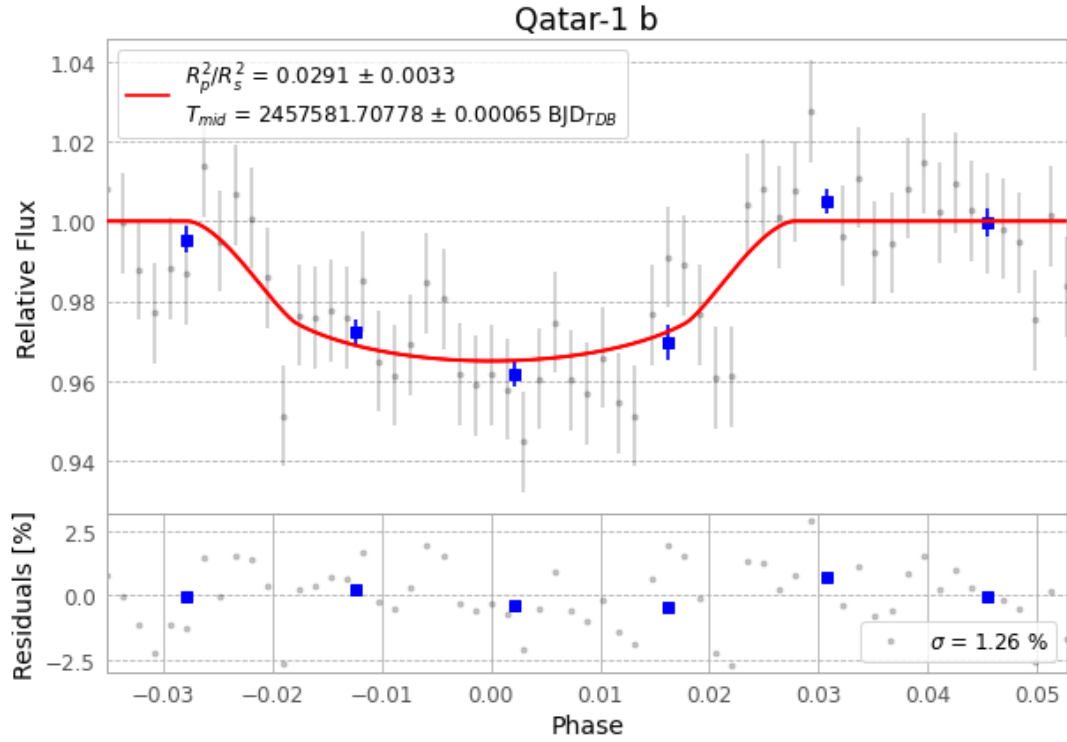


Figure 1 — FinalLightCurve_Qatar-1 b_2016-07-11.png (SHA-256 5b47b5b6586d4bd2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a3c903a6d7b46427...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-1160712034512.FITS → Qatar-1160712064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160712.log (300553b94e0d...), FinalLightCurve_Qatar-1 b_2016-07-11.png (5b47b5b6586d...), FinalLightCurve_Qatar-1 b_2016-07-11.csv (310f87629ed1...)

Compute resources

Wall time 121.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 22:56Z.